

Sheet 3

1) a) $T = 1/F = \frac{1}{24} = 0.041 = 41 \text{ ms}$

B) $T = 1/F = 1/8 = 0.125 \text{ ms}$

C) $T = 1/F = 1/140 = 7.1428 \text{ ms}$

2) a) $F = 1/T = 1/5 = 0.2 \text{ Hz}$

B) $F = 1/T = 1/12 \times 10^6 = 83.3 \text{ KHz}$

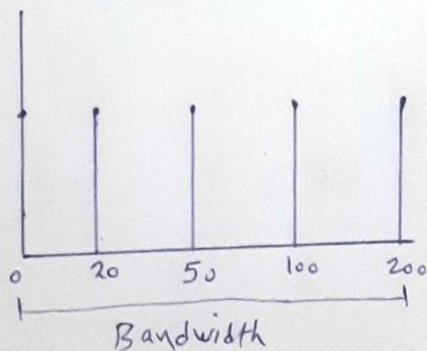
C) $F = \frac{1}{T} = 1/220 \times 10^9 = 4.5 \text{ MHz}$

3) a) $90^\circ \text{ degrees and rad} = 90 \times \frac{2\pi}{360} = \frac{1}{2} \pi \approx 1.5707 \text{ rad}$

B) Max offset $1/1$ cycle then $0 \text{ degrees and rad} = 0$

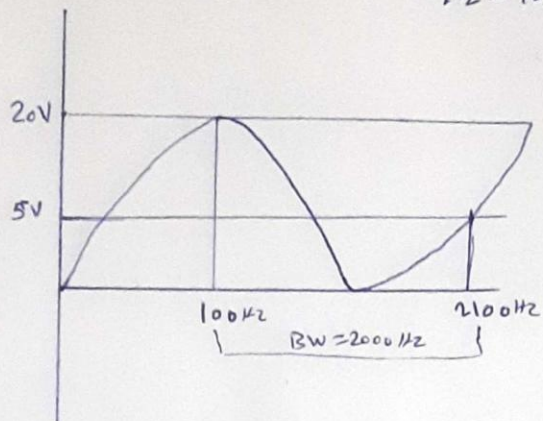
4) Given $0, 20, 50, 100, 200 \text{ Hz}$

$\text{Bandwidth} = F_H - F_L = 200 - 0 = 200$

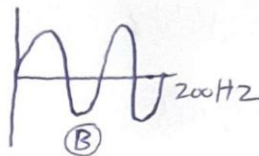
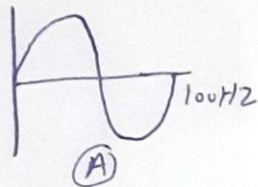


5) $BW = 2000 \text{ Hz}$

$f_L = 100$, $f_H = 2100$



6)



B = has a wider Bandwidth

7) A) 1 bit , 0.001 s $\sim$ Bitrate = $1/0.001 = 1000 \text{ bps} \rightarrow 1 \text{ KBPS}$

B) 1 bit, 2ms $\sim$ Bitrate = $1/2 \times 10^{-3} = 1000 \text{ bps} \rightarrow 1 \text{ KBPS}$

C) 2bit, 20 Ms $\sim$ Bitrate = $2/20 \times 10^{-6} = 500000 \text{ bps} \rightarrow 500 \text{ KBPS}$

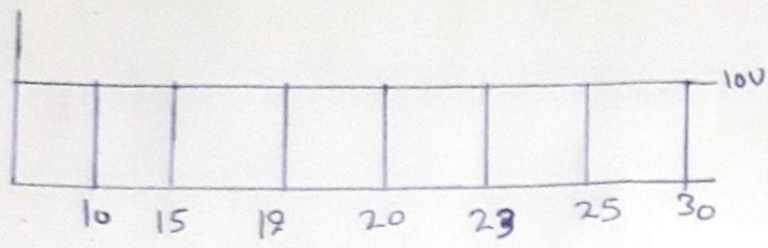
8) a) $= \frac{10}{1000} = 0.01 \text{ s}$

B) $8/1000 = 8 \text{ ms}$

C) $100000 \times 8 = 800000 \text{ KPT} / 1000 = 800 \text{ s}$

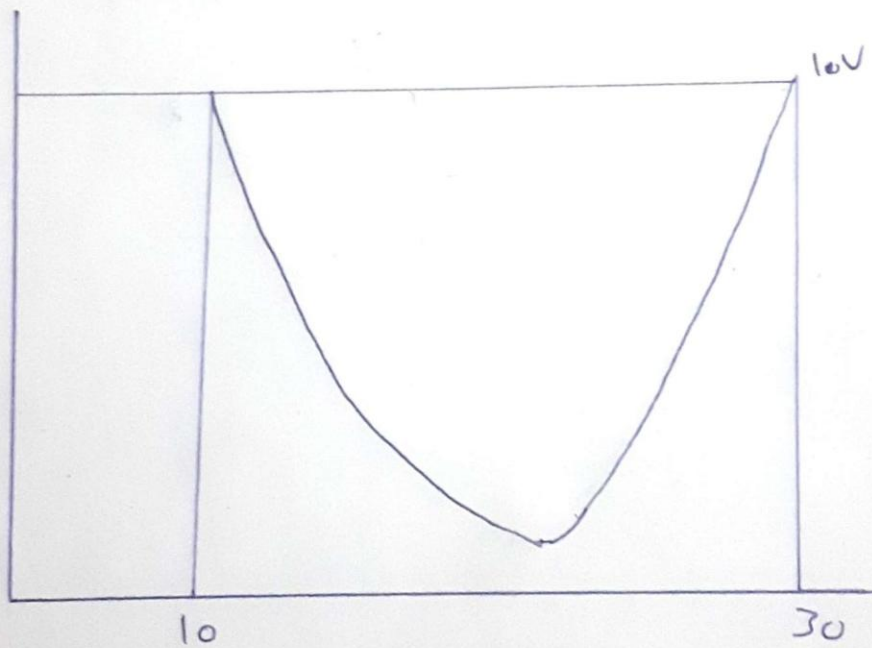
10) $F = \frac{1}{T} = 1/0.004 = 250 \text{ Hz}$

11)

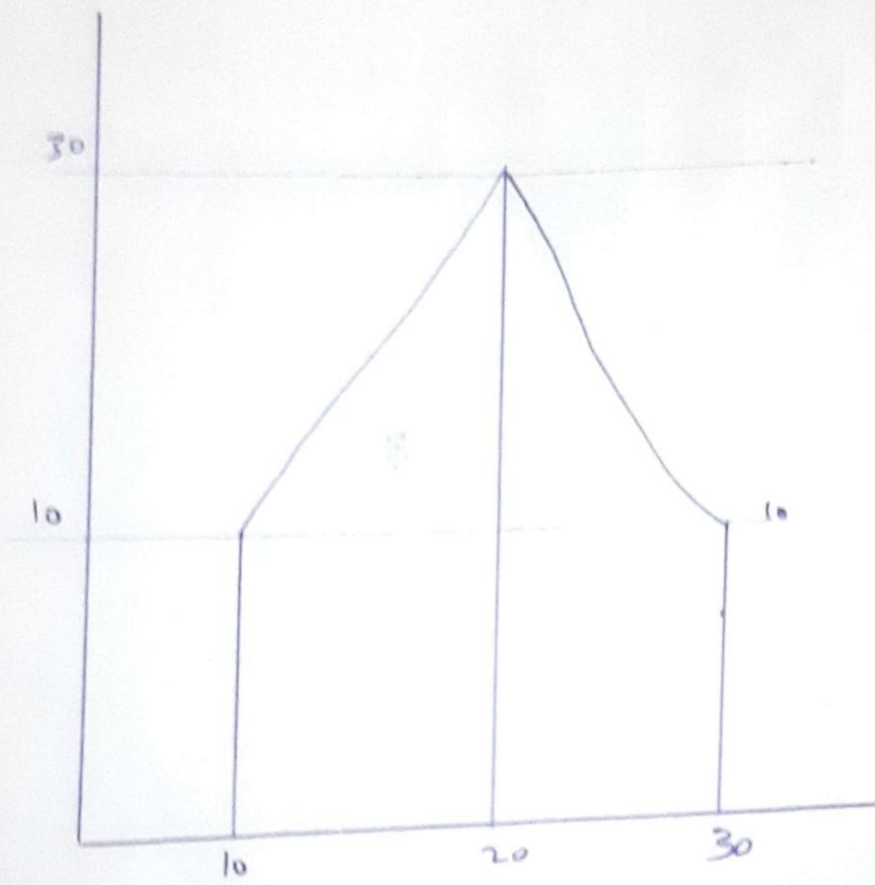


$$B = F_H - F_L = 30 - 10 = 20 \text{ KHz}$$

12)



13)



13. Attenuation in dB

Given: $P_A = 100 \text{ W}$, $P_B = 90 \text{ W}$

Formula: $A(\text{dB}) = 10 * \log_{10}(P_B / P_A)$

$A = 10 * \log_{10}(90 / 100) = -0.458 \text{ dB}$

Answer: -0.458 dB (signal loss)

14. Final Signal Power

Given: Attenuation = -10 dB, $P_{\text{initial}} = 5 \text{ W}$

Formula: $P_{\text{final}} = P_{\text{initial}} * 10^{(A/10)}$

$P = 5 * 10^{(-10/10)} = 0.5 \text{ W}$

Answer: 0.5 W

15. Total Amplifier Gain

Three amplifiers each with +4 dB gain.

Total gain = $4 + 4 + 4 = 12 \text{ dB}$

Linear gain = $10^{(12/10)} = 15.85$

Answer: 12 dB total ($\approx 15.85\times$ amplification)

16. Transmission Time

Bandwidth = 5 Kbps = 5000 bps

Frame = 100,000 bits

Time = bits / rate = $100,000 / 5000 = 20 \text{ s}$

Answer: 20 seconds

17. Maximum Data Rate (Shannon)

SNR = 1000, Bandwidth = 4000 KHz = 4,000,000 Hz

Formula: $C = B * \log_2(1 + \text{SNR})$

$C = 4,000,000 * \log_2(1001) = 39.9 \text{ Mbps}$

Answer: 39.9 Mbps

18. Telephone Line Capacity

Bandwidth = 4 kHz, Signal = 10 V, Noise = 5 mV

$\text{SNR} = (10 / 0.005)^2 = 4,000,000$

$C = 4000 * \log_2(1 + 4,000,000) = 87.7 \text{ kbps}$

Answer: 87.7 kbps

19. File Download Time

File = 2 million bytes = 16 million bits

a) 56 kbps $\rightarrow 16,000,000 / 56,000 = 285.7 \text{ s} \approx 4 \text{ min } 46 \text{ s}$

b) 1 Mbps $\rightarrow 16,000,000 / 1,000,000 = 16 \text{ s}$

Answer: 56 kbps $\rightarrow 285.7 \text{ s}$, 1 Mbps $\rightarrow 16 \text{ s}$

20. Monitor Data Size

Resolution = $1200 \times 1000 = 1,200,000$ pixels

Colors = 1024 $\rightarrow$ bits/pixel = $\log_2(1024) = 10$ bits

Total bits = $1,200,000 \times 10 = 12,000,000$ bits = 12 Mb

Answer: 12,000,000 bits (12 Mb)

21. Theoretical Capacity of a Channel

Formula: $C = B * \log_2(1 + \text{SNR})$

where $\text{SNR} = 10^{(\text{SNRdB} / 10)}$

a. Bandwidth = 20 KHz, SNRdB = 40

$\text{SNR} = 10^{(40/10)} = 10^4 = 10,000$

$C = 20,000 * \log_2(1 + 10,000) = 20,000 * 13.2879 = 265,758 \text{ bps}$

Answer: $\approx 265.8 \text{ Kbps}$

b. Bandwidth = 200 KHz, SNRdB = 4

$\text{SNR} = 10^{(4/10)} = 2.5119$

$C = 200,000 * \log_2(1 + 2.5119) = 200,000 * 1.8122 = 362,449 \text{ bps}$

Answer: $\approx 362.4 \text{ Kbps}$

c. Bandwidth = 1 MHz, SNRdB = 20

$\text{SNR} = 10^{(20/10)} = 100$

$C = 1,000,000 * \log_2(1 + 100) = 1,000,000 * 6.6582 = 6.658 \times 10^6 \text{ bps}$

Answer: $\approx 6.66 \text{ Mbps}$

22. Channel Upgrade (Effect of Doubling Bandwidth or SNR)

a. If we double the bandwidth:

Since $C = B * \log_2(1 + \text{SNR})$, doubling B doubles the data rate.

Answer: Capacity is doubled.

b. If we double the SNR:

$C' = B * \log_2(1 + 2 * \text{SNR})$

Improvement ratio = $\log_2(1 + 2 * \text{SNR}) / \log_2(1 + \text{SNR})$

Doubling SNR increases capacity but not by double (diminishing returns).

Answer: Capacity increases, but less than 2x.

23. Total Delay (Latency)

Given:

- Frame size = 5,000,000 bits
- 10 routers: each has 2 μs queuing + 1 μs processing
- Distance = 2000 km = 2,000,000 m
- Propagation speed = $2 \times 10^8 \text{ m/s}$
- Bandwidth = 5 Mbps = 5,000,000 bps

Step 1: Transmission delay = $5,000,000 / 5,000,000 = 1.0 \text{ s}$

Step 2: Propagation delay = $2,000,000 / (2 \times 10^8) = 0.01 \text{ s} = 10 \text{ ms}$

Step 3: Queuing + Processing = $10 \times (2 + 1) \mu\text{s} = 30 \mu\text{s} = 0.00003 \text{ s}$

Total delay = $1.0 + 0.01 + 0.00003 = 1.01003 \text{ s}$

Answer: $\approx 1.01003 \text{ seconds total delay}$